

PETIO Firmware-Derived Bill of Materials (BOM)

Generated on: 2026-05-30

Source: d:\PETio\controller\firmware\Firmware\Firmware.ino

Schematics Cross-Verification:

Found in repo: False

Note: No schematics/PCB source files were found in the repository. Pin mapping verification

== Microcontrollers & Wireless ==

Component Name: Espressif ESP8266EX (ESP8266-based module; exact module variant not specified)

Operating Voltage: 2.5?3.6 V (nominal 3.3 V). Firmware drives GPIO directly; assumes 3.3 V logic

Current Consumption: Firmware does not implement sleep; expect Wi-Fi active/idle currents per

Communication Protocol: UART0 (Serial debug) @ 115200 baud; IEEE 802.11 b/g/n 2.4 GHz Wi-Fi;

Datasheet Link: https://documentation.espressif.com/0a-esp8266ex_datasheet_en.html

Firmware Evidence:

- Firmware.ino:L12-L26 (ESP8266 libraries included)
- Firmware.ino:L564 (Serial.begin(115200))
- Firmware.ino:L405-L430 (ESP8266WebServer(80), DNSServer, dnsServer.start(53, ...))
- Firmware.ino:L499 (WiFiClientSecure.setInsecure())
- Firmware.ino:L536-L538 (NTPClient + pool.ntp.org parameters)

Discrepancy Flags:

- TLS certificate validation is disabled (setInsecure) which is a security risk, not a data

== Actuators ==

Component Name: Tower Pro MG996R (as referenced by firmware comment) - high-torque servo motor

Operating Voltage: 4.8?6.6 V (per manufacturer product spec). Control signal is PWM from ESP8266

Current Consumption: Firmware does not calculate motor current. Manufacturer specs list idle

Communication Protocol: Servo PWM control (Servo.h). Typical 50 Hz pulse train; firmware sets

Datasheet Link: <https://towerpro.com.tw/product/mg996r/>

Firmware Evidence:

- Firmware.ino:L7-L9 (MG996R servo referenced in header comment)
- Firmware.ino:L36-L81 (SERVO_* pulse definitions)
- Firmware.ino:L264-L267 (Servo.attach + writeMicroseconds usage)
- Firmware.ino:L279-L327 (feeding control + ramping logic)

Discrepancy Flags:

- Firmware assumes continuous rotation behavior (feeding) while MG996R is commonly an angle

== User Input (Switches) ==

Component Name: Momentary tactile push button switch (generic; exact part number not specified)

Operating Voltage: 3.3 V logic (GPIO input with internal pull-up enabled).

Current Consumption: Negligible; GPIO pull-up current depends on internal pull-up and switch

Communication Protocol: GPIO digital input (active-low with debouncing in firmware).

Datasheet Link:

Firmware Evidence:

- Firmware.ino:L37-L41 (FEED_NOW_PIN=5, BUTTON_PIN=FEED_NOW_PIN)
- Firmware.ino:L594-L595 (pinMode(BUTTON_PIN, INPUT_PULLUP))
- Firmware.ino:L104-L113 (debounce constants)

Discrepancy Flags:

- Exact switch manufacturer/part number not specified in firmware; datasheet link cannot be

Component Name: Momentary tactile push button switch (generic; exact part number not specified)

Operating Voltage: 3.3 V logic (GPIO input with internal pull-up enabled).

Current Consumption: Negligible; GPIO pull-up current depends on internal pull-up and switch

Communication Protocol: GPIO digital input (active-low).

Datasheet Link:

Firmware Evidence:

- Firmware.ino:L38-L39 (ADD_PORTION_PIN=4)
- Firmware.ino:L592-L593 (pinMode(ADD_PORTION_PIN, INPUT_PULLUP))

Discrepancy Flags:

- Exact switch manufacturer/part number not specified in firmware; datasheet link cannot be

Component Name: Momentary tactile push button switch (generic; exact part number not specified)

Operating Voltage: 3.3 V logic (GPIO input with internal pull-up enabled).

Current Consumption: Negligible; GPIO pull-up current depends on internal pull-up and switch

Communication Protocol: GPIO digital input (active-low).

Datasheet Link:

Firmware Evidence:

- Firmware.ino:L39-L40 (REDUCE_PORTION_PIN=12)
- Firmware.ino:L592-L594 (pinMode(REDUCE_PORTION_PIN, INPUT_PULLUP))

Discrepancy Flags:

- Exact switch manufacturer/part number not specified in firmware; datasheet link cannot be

== Indicators (LEDs) ==

Component Name: Red LED indicator (generic; exact part number not specified) - status LED (RED)

Operating Voltage: Driven by ESP8266 GPIO at 3.3 V. LED forward voltage depends on the chosen

Current Consumption: Depends on LED + series resistor. Firmware drives GPIO HIGH/LOW; typical

Communication Protocol: GPIO digital output.

Datasheet Link:

Firmware Evidence:

- Firmware.ino:L31-L34 (RED_LED_PIN=13)
- Firmware.ino:L201-L207 (pinMode OUTPUT and setState logic)

Discrepancy Flags:

- LED part number and series resistor value not specified in firmware; datasheet link cannot be

Component Name: Blue LED indicator (generic; exact part number not specified) - status LED (BLUE)

Operating Voltage: Driven by ESP8266 GPIO at 3.3 V. LED forward voltage depends on the chosen

Current Consumption: Depends on LED + series resistor. Firmware drives GPIO HIGH/LOW; typical

Communication Protocol: GPIO digital output.

Datasheet Link:

Firmware Evidence:

- Firmware.ino:L31-L34 (BLUE_LED_PIN=16)
- Firmware.ino:L201-L209 (pinMode OUTPUT and setState logic)

Discrepancy Flags:

- LED part number and series resistor value not specified in firmware; datasheet link cannot be

Validation:

- UART baud rate is explicitly configured in firmware (115200).
- Captive portal ports are explicitly configured (HTTP server 80; DNS 53).
- NTP server host and update interval are explicitly configured (pool.ntp.org; 3600000 ms).
- Servo control pulses are explicitly configured (microsecond pulse widths).

Known Gaps Requiring Schematics:

- Exact ESP8266 module variant (ESP-12E/ESP-12F/NodeMCU/etc.) is not specified in firmware.
- Exact LED part numbers and their series resistor values are not specified in firmware.
- Exact button switch part numbers are not specified in firmware.
- Power supply/regulator selection and servo power rail design are not represented in firmw